# PYTHON WINTER SPRINT 2026

## Core Problem of 7th Jan 2026 : Clinical Data Auditer

Each Day have one constant theme and three different tracks : Foundation Track especially due to W3B Collaboration, Core Track for those following since DAY-1 and the Project Track for those who have their basics of Python Cleared, they will make these end-to-end projects

## # FOUNDATION TRACK (FOR ABSOLUTE BEGINNERS AND NEW JOINERS

[Expected Time to Finish – 20 minutes]

Resource: https://www.youtube.com/watch?v=UrsmFxEIp5k from start till 1hr 40 minutes

Core Topic: Variables, Data Types, Strings

🧠 Problem Statement

Write a Python program that:

1. Takes the following input from the user:
   - Patient Name
   - Age
   - Heart Rate
   - City
2. Perform simple string operations:
   - Convert patient name to Title Case
   - Convert city name to Uppercase
3. Print a clean Clinical Data Record in a formatted way.

⚠️ This task is only about recording data, not validating or auditing it.

📌 Submission Guidelines

Submit Before 11 pm 7th January

GitHub repository is mandatory - Commit history should reflect real work

Optionally, Include a short README explaining: What the system does, Audit logic and Disclaimer

Sample Input:

```
Enter patient name: ali khan
Enter age: 21
Enter heart rate: 78
Enter city: delhi
```

Sample Output:

```
--------------------------------
   CLINICAL DATA RECORD
--------------------------------
Patient Name : Ali Khan
Age          : 21
Heart Rate   : 78 bpm
City         : DELHI
--------------------------------
Note: This record is for data entry only.
```

Beginners Please ask in Group if you get to encounter any problem, any kind of problem in this, or just DM me

Every One was once a beginner !!

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# # CORE TRACK (FOR STUDENTS WHO HAVE BEEN LEARNING FROM DAY 1)

Core Topic: Functions and Recursion

Resource: https://www.youtube.com/watch?v=UrsmFxEIp5k from 4hr 50 minutes till 5hr 54 minutes

[Expected Time to Finish - 30 minutes]

🧠 **Problem Statement**

Write a Python program that audits basic clinical data using functions and recursion.

The program should:

Take patient details:

- Name
- Age
- Heart rate
- Number of vitals readings

Use functions to:

- Validate age
- Validate heart rate
- Decide audit status

Use recursion to:

- Take multiple heart rate readings (one by one)

Audit Logic:

- Age must be between 0–120
- Heart rate must be numeric
- If any heart rate > 180 → Warning
- If invalid input → Flag

Final Audit Status:

- FAIL → Any flag
- REVIEW → No flags, but warnings
- PASS → Clean data

📌 Submission Guidelines

Submit Before 11 pm 7th January

GitHub repository is mandatory - Commit history should reflect real work

Optionally, Include a short README explaining: What the system does, Audit logic and Disclaimer

Sample Input:

```
Enter patient name: Ayaan Khan
Enter age: 22
Enter number of heart rate readings: 3

Enter heart rate reading: 76
Enter heart rate reading: 185
Enter heart rate reading: 82
```

Sample Output:

```
--- CLINICAL AUDIT RESULT ---
Patient : Ayaan Khan
Status  : REVIEW
Flags   : None
Warnings: ['High heart rate detected']
```

Sample Input (With Flag Case)

```
Enter patient name: Sara Ali
Enter age: twenty
Enter number of heart rate readings: 2

Enter heart rate reading: 72
Enter heart rate reading: 80
```

```
--- CLINICAL AUDIT RESULT ---
Patient : Sara Ali
Status  : FAIL
Flags   : ['Invalid age']
Warnings: None
```

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## # PROJECT TRACK (FOR ADVANCED PARTICIPANTS, MINI PROJECT)

[Expected Time to Finish – 3-4 hours]

🩺 Clinical Data Audit System

Project Instructions (Advanced Track)

### 🎯 Objective

Build a CLI-based Clinical Data Audit System that audits patient-provided clinical data for safety, validity, and consistency. The system must be non-diagnostic and focus only on data auditing, similar to backend validation layers used in real healthcare software.

### 📥 User Inputs

Your program must collect the following:

- Patient Name
- Age
- Heart Rate (bpm)
- Blood Pressure:
  - Systolic
  - Diastolic
- Allergy information (yes / no)

> 📌 Submission Guidelines
> Submit Before 11 pm 7th January
> GitHub repository is mandatory - Commit history should reflect real work
> Optionally, Include a short README explaining: What the system does, Audit logic and Disclaimer

### 🔍 Audit Rules (Mandatory)

**VALIDATION RULES**

- Age must be a number and within 0–120
- Heart rate must be numeric
- Blood pressure values must be numeric
- Systolic BP must be greater than diastolic BP
- Allergy field must be either yes or no

**SAFETY RULES (WARNINGS)**

- Heart rate outside 40–180 bpm → Warning
- Systolic BP outside 70–200 → Warning
- Diastolic BP outside 40–130 → Warning

### 🧾 Audit Classification

Based on audit results, assign one status:

- PASS → No flags or warnings
- REVIEW → Warnings present but no hard failures
- FAIL → Any validation rule violated

### 🔑 Traceability Requirements

- Generate a random unique Audit ID for every audit for further traceability
- Attach the Audit ID to:
  - Console output
  - File logs

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## # PROJECT TRACK (FOR ADVANCED, SYSTEM-LEVEL PROBLEM SOLVING)

### 💾 File I/O Requirements

Create an append-only audit log file, Each audit record must include: Audit ID, Timestamp, Patient name, Audit status (Pass/Review/Fail), Flags (if any validation rule fails), Warnings (if any)

- Older audit records must never be overwritten

### 🖨️ Output Requirements

Display a clean Audit Report in the console containing:

1. Audit ID
2. Timestamp
3. Patient name
4. Audit status (Pass/Review/Fail)
5. Flags (Failure of Validation Rules)
6. Warnings
7. Disclaimer that this is a non diagnostic Report Only, Not medically certified

⚠️ Strict Constraints

❌ No diagnosis

❌ No treatment recommendations

❌ No medical advice

✅ Audit & validation only

Sample Run:

▶ Program Start

```yaml
=== Clinical Data Audit System ===
Enter patient name: arshad ali
Enter age: 22
Enter heart rate (bpm): 185
Enter systolic BP: 118
Enter diastolic BP: 76
Any known allergies? (yes/no): no
```

🖥 Console Output

```yaml
------------------------------------
        CLINICAL AUDIT REPORT
------------------------------------
Audit ID          : AUD-483921
Timestamp         : 2026-01-07 18:42:11
Patient Name      : Arshad Ali
Audit Status      : REVIEW

Audit Flags:
None

Warnings:
Heart rate outside typical safe range

Audit record saved for traceability.
This system performs data audit only.
No diagnosis or medical advice provided.
------------------------------------
```

📁 Audit Log File (`clinical_audit_log.txt`)

```yaml
--------------------------------------
Audit ID          : AUD-483921
Timestamp         : 2026-01-07 18:42:11
Patient Name      : Arshad Ali
Audit Status      : REVIEW
Audit Flags: None
Warnings:
- Heart rate outside typical safe range
Note: Non-diagnostic clinical data audit
--------------------------------------
```

🧪 Sample Run – FAIL Case

▶ Input

```yaml
Enter patient name: sara khan
Enter age: two
Enter heart rate (bpm): 78
Enter systolic BP: 110
Enter diastolic BP: 80
Any known allergies? (yes/no): yes
```

🖥 Output

```yaml
Audit Status      : FAIL

Audit Flags:
Invalid age format

Warnings:
None
```